



Did those with higher education change their political trust more than more than those with lower education as a result of the economic recession?

In submitting this assignment:

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1. EXECUTIVE SUMMARY

Our research aims at investigating the relationship between higher or lower level of education and political trust as a result of the Economic Recession. Our hypothesis is that given their greater capacity to properly assess the governments' response to the crisis, coupled with their stronger commitment to democratic values, we hypothesize that higher educated citizens should react differently in terms of political trust from those with less education. We believe that a crisis such as the Great Recession undermines people's trust and confidence in the actors and institutions of democratic government. Thus, our research is focused on the direct effects of a significant external event which has steadily increased the perceived political risk by individuals on political trust.

In order to properly analyze the impact of the Economic Recession happened in 2007, the ESS round 3 – 2006 and ESS round 5 – 2010 have been selected, which can provide an insight on the situation before and after the crisis.

2. INTRODUCTION

The investigation into whether individuals with higher levels of education experienced more pronounced shifts in political trust compared to their less-educated counterparts in the aftermath of the economic recession stands as a crucial exploration of the very foundations of social-economics dynamics. Within the framework of this applied research project, our goal is to examine the potential influence of education on an individual's perspective regarding political trust. This endeavor is of paramount importance, as it is imperative to gain a profound understanding of the intricate relationship between these two variables to effectively address this global challenge. Our approach will involve conducting a comprehensive multivariate analysis of pertinent data extracted from the European Social Survey (ESS) Rounds 3 and 4, with our units of analysis being citizens of different level of education. Furthermore, it is essential to underscore the significance of this study, as it has the potential to shed light on the far-reaching consequences of an economic shock, such as the one experienced during the Economic Recession, on the political stability of Europe. In this report, we have chosen to focus our analysis on the variables of education and political trust to determine whether the change in political trust among highly educated individuals is more substantial than that among less-educated individuals. An underlying assumption guiding our research is that, generally, individuals with higher levels of education may exhibit greater skepticism towards government and its actions.

3. BACKGROUND

Trust is the bedrock of functioning societies, fostering cooperation, cohesion, and economic growth. Globally, trust in public institutions is instrumental in advancing Sustainable Development Goals (SDGs) spanning poverty reduction, climate action, and inclusive societies. Notably, the pivotal role of trust was evident during crises like the COVID-19 pandemic, where higher public trust correlated with better outcomes. Yet, a growing concern surrounds the erosion of trust, giving rise to extreme political views, protests, and conflicts. Measuring trust through surveys reveals variations across countries, with higher-income nations generally displaying more institutional trust. Nevertheless, recent decades have witnessed declining trust, partly due to factors like economic insecurity and perceived government ineffectiveness. Education also shapes perceptions of political systems. This decline is closely tied to economic insecurity, as citizens expect governments to provide security and quality services. When governments falter, trust erodes, hindering governance and service provision. Political disengagement and misinformation exacerbate this challenge. Addressing economic insecurity and improving public institutions' effectiveness and accountability are interlinked priorities. Governments meeting citizen needs and providing security are more likely to rebuild trust. Rebuilding trust is a long-term endeavor requiring consistent investment in bridging social, economic, and political divisions. Policies promoting social protection, citizen participation, transparency, and leveraging technology are vital steps in restoring trust in public institutions.

4. DATA DESCRIPTION

This research uses data from the European Social Survey (ESS), which since its inception in 2001, a rigorous cross-national survey with a strong academic focus has been carried out throughout Europe. This survey involves conducting face-to-face interviews with newly chosen, representative samples every two years. In order to conduct our analysis we gathered data from the ESS round 3 which was conducted in 2006 and ESS round 4 which was conducted in 2010 and we have appended them. Since round 3 does not include Romania and Latvia while round 4 does not include Austria, in order to compare the two periods we have removed those countries, before starting our analysis. We opted for ESS data for 2010 instead of 2008 due to the fact that long-term changes in behavior would be more prominent instead of rash and uncommon reactions to the crisis in the heat of the

moment in 2008, moreover even though the recession started in December 2007 and continued throughout 2009 the 2008 dataset doesn't fully capture the result of such crisis. The complex multistage probability sampling method that has been used in our two data sets, is a rigorous and standardized method where trained interviewers conduct in-person interviews with people from selected households. Within each selected cluster (cluster analysis) households and individuals selected from each household are randomly chosen. The quality of the data, in terms of reliability, validity, accuracy and precision, is ensured by interviewer training, supervision and monitoring which is thoroughly carried out by the ESS. It is fundamental to specify that our analysis has been conducted using weights. We have created a new weight called overallwght that is equal to the product of pspwght, correcting for the stratification sampling design at the country level and pweight, accounting for the over-sampling of some countries. The dependent variable '*Trust in political parties*' is a categorical ordinal variable that represents the trust citizens have in political parties as a direct relationship between citizens and political parties. It is normally distributed and has a total absolute frequency of 140,966 observations. It has been measured through an anchored scale with a range from 0 to 10, where the value 0 designates '*No trust at all*' and value 10 stands for '*Complete trust*'. As shown in (**Exhibit 1**). While '*Refusal*', '*Don't know*' and '*No answer*' are regarded as Missing values.

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Trust in political parties	Freq.	Percent	Cum.	Value	Category
No trust at all	25,013	17.74	17.74	0	Not possible to harmonise into 5-level ISCED
1	14,417	10.23	27.97	1	Less than lower secondary education (ISCED 0-1)
2	17,552	12.45	40.42	2	Lower secondary education completed (ISCED 2)
3	19,118	13.56	53.98	3	Upper secondary education completed (ISCED 3)
4	16,068	11.40	65.38	4	Post-secondary non-tertiary education completed (ISCED 4)
5	23,834	16.91	82.29	5	Tertiary education completed (ISCED 5-6)
6	11,473	8.14	90.43	55	Other
7	8,407	5.96	96.39	77	Refusal*
8	3,496	2.48	98.87	88	Don't know*
9	895	0.63	99.51	99	No answer*
Complete trust	693	0.49	100.00		
Total	140,966	100.00			

*) Missing Value

Exhibit 1: Table of 'trstprt' variable

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Years of full-time education completed	Freq.	Percent	Cum.	Years of full-time education completed	Freq.	Percent	Cum.
0	838	0.60	0.60	0	838	0.60	0.60
1	283	0.20	0.80	1	283	0.20	0.80
2	420	0.30	1.11	2	420	0.30	1.11
3	1,075	0.77	1.88	3	1,075	0.77	1.88
4	3,208	2.30	4.18	4	3,208	2.30	4.18
5	1,046	0.75	4.93	5	1,046	0.75	4.93
6	3,725	2.67	7.60	6	3,725	2.67	7.60
7	3,257	2.34	9.94	7	3,257	2.34	9.94
8	7,966	5.72	15.66	8	7,966	5.72	15.66
9	7,903	5.67	21.33	9	7,903	5.67	21.33
10	10,111	7.26	28.58	10	10,111	7.26	28.58
11	14,780	10.61	39.19	11	14,780	10.61	39.19
12	1	0.00	39.19	12	1	0.00	39.19
12	23,481	16.85	56.04	12	23,481	16.85	56.04
13	12,512	8.98	65.01	13	12,512	8.98	65.01
14	10,003	7.18	72.19	14	10,003	7.18	72.19
15	10,042	7.21	79.40	15	10,042	7.21	79.40
16	8,994	6.45	85.85	16	8,994	6.45	85.85
17	6,750	4.84	90.69	17	6,750	4.84	90.69
18	5,222	3.75	94.44	18	5,222	3.75	94.44
19	1	0.00	94.44	19	1	0.00	94.44
19	2,663	1.91	96.35	19	2,663	1.91	96.35
20	2,464	1.77	98.12	20	2,464	1.77	98.12

Exhibit 2: Table of 'eduyrs' variable

5. MULTIVARIATE ANALYSIS

In the pursuit of addressing our research inquiry, it becomes incumbent to conduct a meticulous analysis of the variables at our disposal. Within the context of **data screening**, a paramount objective is to ascertain the presence of missing values and outliers.

Missing values, within the framework of statistical analysis, pertain to data points or observations that are conspicuously absent or exhibit incompleteness within a dataset. The scrutiny of missing values assumes a pivotal role in data analysis, as the presence of such

Our explanatory variable, 'years of full-time education completed' is a numerical interval variable which answers to the question about how many years of education respondents have completed, whether full-time or part-time, reported in full-time equivalents and includes compulsory years of schooling. It is also normally distributed and has a total absolute frequency of 139,364 observation. As shown in (**Exhibit 2**)

gaps has the potential to significantly impact the precision and reliability of statistical findings.

In this specific dataset, an inspection of the dependent variable, denoting "*Political trust*", reveals the existence of 4545 missing values, constituting 3.12% of the dataset. On account of the relative insignificance of this proportion in relation to the total number of observations, it is judicious to assert that these missing values may be judiciously omitted, as their relevance is deemed nominal. This determination is predicated upon the examination of the proportion of missing values vis-à-vis the entire dataset, which conclusively renders their elimination as a safe course of action. A comparable scrutiny was executed on the independent variable representing "education," wherein it was discerned that 1602 missing values, corresponding to 1.14% of all responses, were extant. Thus, a similar decision to eliminate these missing values was made on grounds of their immateriality. Outliers, characterized as anomalous data points that exhibit conspicuously elevated values in relation to their counterparts, warrant a distinct consideration. In the context of "political trust," a thorough examination indicates the absence of outliers, a conclusion substantiated by the analysis of a boxplot (Exhibit 3), which unequivocally ascertains their non-existence. As a consequence, it is deemed secure to retain the entirety of the data pertaining to this variable.

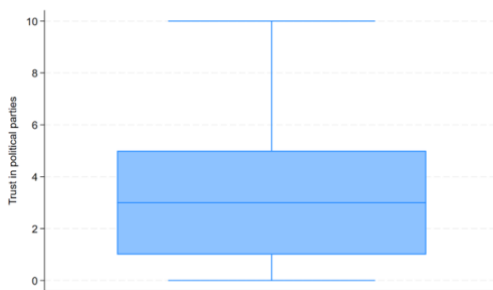


Exhibit 3: Boxplot of 'trstprt' variable

Conversely, upon an analogous inspection of the independent variable encompassing "education," the results derived from the boxplot (Exhibit 4) incontrovertibly demonstrate the presence of numerous outliers. Accordingly, the decision was taken to eliminate 1442 values. This choice is anchored in the definition of outliers as observations that surpass a threshold equivalent to the mean plus or minus three times the standard deviation.

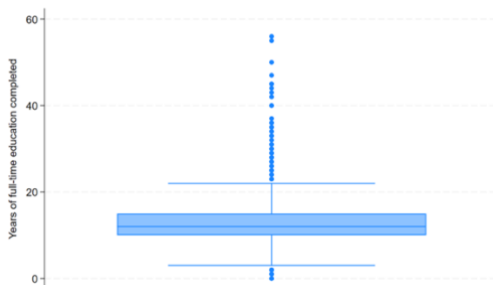


Exhibit 4: Boxplot of 'edyurs' variable

Subsequent to this initial phase of data screening, it is imperative to gauge whether the variables adhere to a normal distribution. To undertake this appraisal, a dual-pronged approach encompassing both graphical and statistical methodologies was adopted. The graphical approach entailed the presentation of histograms for both variables, while statistical parameters, namely skewness and kurtosis, were employed as quantitative measures.

Skewness, elucidating the symmetry of the distribution, is expected to manifest values within the range of -1 to 1. In contrast, kurtosis, explicating the thickness of distribution tails and the prevalence of extreme observations, is anticipated to manifest values ranging from 2 to 4 in the context of a normal distribution. Upon a meticulous evaluation, it is discerned that the independent variable explicating "education" indeed adheres to a normal distribution, as substantiated by the associated histogram (Exhibit 5). Furthermore, the respective kurtosis and skewness values, quantified at 3.153067 and -0.0399829, align harmoniously with the parameters indicative of a normal distribution.

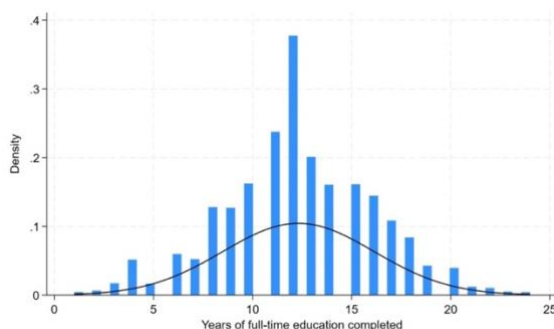
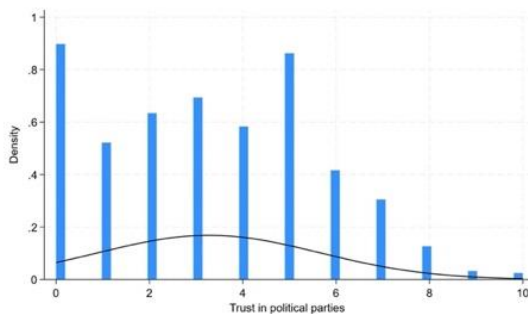


Exhibit 5: Histogram of 'edyurs' variable

In a similar vein, the dependent variable elucidating "political trust" is revealed to exhibit a normal distribution, a deduction substantiated by the representation of its histogram (Exhibit 6). The calculated kurtosis value of 2.222836 and the skewness value of 0.246819 attune faithfully to the attributes concomitant with a normal distribution.



Having checked the appropriateness of our data, we can move on to analyse the relationship of the 2007 crisis to political trust based on education level. To tell whether people with higher education and lower education changed their political trust to different extents, we first created two dummy variables. The first was to check whether the person has high or low education, here we decided that high education was someone with above average (>12.3) years of education.

Exhibit 6: Histogram of 'eduysr' variable

Accordingly, if a person has had more than 12.3 years of education, the variable has a value of 1 and 0 otherwise. To check whether the response is from the 2006 or 2010 round we created another dummy variable, where the value is 0 if it's from the 2006 round and 1 if it's from the 2010 round. With this, we can compare the change in the average political trust among low-educated people after the crisis, as well as the change in the average political trust among highly educated people after the crisis. To do this we regressed political trust by the round, in each case considering only highly or lowly educated people. The results of the two regressions show us, that the relationship between the change in political opinion and whether it was after the crisis was significant in both cases, highly educated and lowly educated. This was determined by the Prob > F statistic being 0.0000 (low education) and 0.0002 (high education) respectively. Additionally, we can see that the $P > |t|$ is 0.000 in both cases. By analysing the coefficients of both regressions, we can see that in the case of highly educated people, it changed by -0.1526451 , whereas it only changed by -0.0678158 for lower-educated people. This difference in change is also visible in the boxplot graphs of the two education levels between the years, where the mean for the highly educated people is visibly much lower (Exhibit 8) than the new mean for lower-educated people (Exhibit 7).

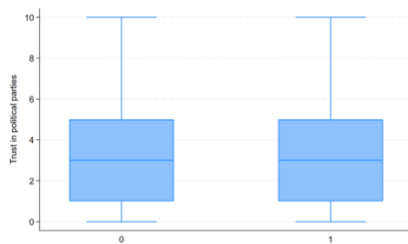


Exhibit 6 - Low educated people

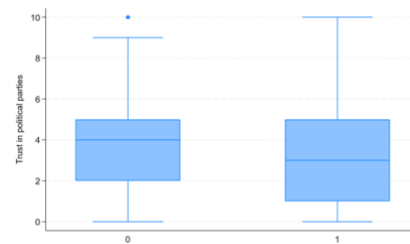


Exhibit 7 – High educated people

6. CONCLUSION

To conclude, through a detailed-oriented analysis using regression analysis, the result shows that the model is statistically significant and that there is a relationship between years of education and the change in political trust following the crisis of 2007. As a result, individuals with a high level of education changed their opinions on politics more drastically compared to those with a lower level of education, after such external event. This is significant as it shows that highly educated people might be more aware of political actions and their impact on the economy. As a result, politicians and business leaders might need to develop more sophisticated solutions to problems in order to appeal to people with higher education.

7. REFERENCES:

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